AgriAgent

An Agentic Multimodal Framework for Adaptive Agricultural Decision Support Using Satellite and Weather Intelligence

Capstone Project – M.Tech AI-ML

Team Members

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Get Started

Problem Statement

Agricultural decision-making often depends on information from multiple sources such as satellite imagery, weather conditions and agricultural knowledge. However, these sources are usually analyzed separately, making it difficult to obtain timely and actionable insights.

AgriAgent proposes an Agentic AI system that autonomously integrates multimodal agricultural information to assess crop conditions, identify risks and provide adaptive decision support to farmers.

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Why is this Problem Important?

🌱 Crop Health

🛰️ Satellite Intelligence

🌦️ Weather Intelligence

💧 Resource Optimization

🤖 Agentic Decision Support

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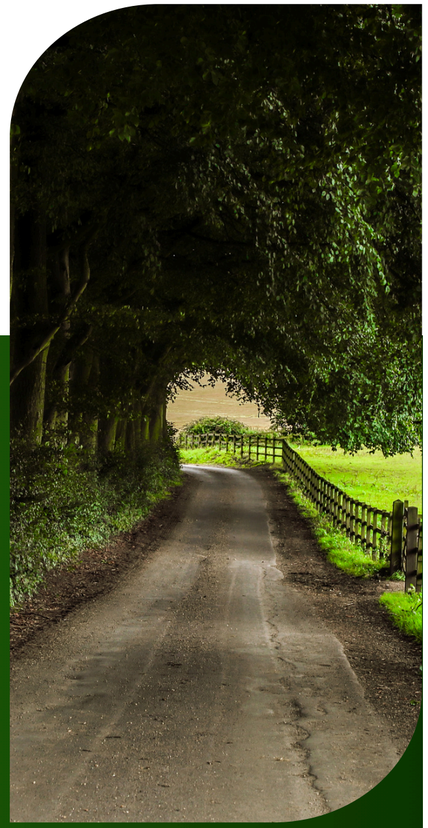
Expected Outcomes & Initial Scope



The proposed AgriAgent system is expected to:

- Analyze **satellite-derived vegetation indicators such as NDVI.**
- Integrate **historical and forecast weather information.**
- Assess **crop/vegetation health and environmental conditions.**
- Identify potential **agricultural risks**.
- Generate **adaptive recommendations**, such as irrigation or crop-management actions.
- Provide **explainable recommendations** supported by relevant data and agricultural knowledge.

Inputs:

🛰️ Satellite Data + 🌦️ Weather Data + 📍 Farm Location

⬇️

Processing:

Satellite Analysis + Weather Analysis + Risk Assessment + Agentic Reasoning

⬇️

Outputs:

🌱 Crop/Vegetation Health Assessment
⚠️ Risk Assessment
💧 Irrigation Recommendation
📋 Crop-Management Recommendation

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Technology & Data Requirements

Technology

Python | LLM | LangGraph/LangChain | ML | RAG | Geospatial Processing | Weather API | Streamlit

Data Sources

Sentinel-2/Landsat | Weather Data | Agricultural Datasets | Agricultural Knowledge Documents

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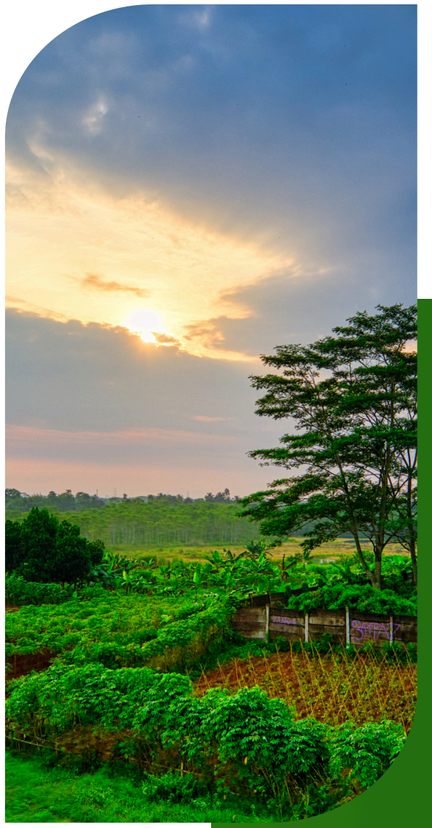
Research & Evaluation

Research Objective

To investigate whether an agentic multimodal approach can improve the ability to combine satellite and weather information for adaptive agricultural decision support.

Evaluation

- NDVI/vegetation assessment accuracy
- ML prediction performance
- Recommendation relevance
- Agent/tool selection performance
- Response accuracy and explainability



Research Potential

The project can potentially contribute toward:

CAPSTONE → MONOGRAPH → CONFERENCE/RESEARCH PUBLICATION

THANK YOU

AgriAgent

Integrating Satellite Intelligence, Weather Data & Agentic AI for Smarter Agriculture

QUESTIONS & DISCUSSION

We look forward to your guidance and feedback